PLANT RECOGNITION

Computer Vision for Precision Agriculture

Deep Learning-Based Plant Species Identification and Disease Detection Using Transfer Learning with VGG16

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Abstract

Traditional plant disease management relies on broad-spectrum pesticide application — a practice that is chemically wasteful, environmentally damaging, and economically inefficient. This paper presents the development and evaluation of an AI-driven computer vision system for real-time plant species identification and disease detection, designed for integration into precision agriculture platforms and robotic systems.

We systematically evaluated a pipeline of machine learning and deep learning models against the PlantVillage Dataset (~55,000 images, 38 plant health classes). Classical machine learning approaches — including Random Forest with PCA, HOG+Decision Tree, LBP+k-NN, Gabor+LDA+Logistic Regression, and SVC — achieved peak accuracies of 50–73.6%. Transfer learning with VGG16 (pre-trained on ImageNet), augmented with dataset segmentation, mixed precision training, and Grad-CAM interpretability analysis, achieved a test accuracy of 94–96% and an F1-score of 95%.

The primary contributions of this work are: (1) a rigorous comparative benchmark of classical vs. deep learning approaches on a large multi-class agricultural image dataset; (2) empirical demonstration that background segmentation and class-balanced augmentation are the highest-leverage preprocessing steps; (3) use of Grad-CAM to validate that the final model attends to disease-relevant leaf regions rather than background artefacts; and (4) an open-source implementation suitable for deployment on mobile devices and robotic platforms for field use.

1. Introduction

Agricultural productivity depends critically on early, accurate detection of plant disease. The Food and Agriculture Organization estimates that plant diseases cause annual crop losses of 20–40% globally, with the economic impact disproportionately affecting smallholder farmers in developing regions. Conventional disease management — aerial pesticide application, manual inspection, and reactive treatment — is both imprecise and costly, generating unnecessary chemical exposure for both ecosystems and farm workers.

Computer vision and deep learning offer a pathway to precision disease detection: a camera-equipped device (smartphone, drone, or ground robot) captures an image of a leaf, and a trained classifier identifies the species and disease state in real time. This enables targeted, spot-treatment interventions rather than blanket chemical application — reducing input costs, environmental load, and human exposure simultaneously.

The principal challenge in this domain is not the availability of models (convolutional neural networks have been well-studied for image classification since 2012) but the practical pipeline: which datasets are appropriate, how to address class imbalance and domain shift, how to validate that a model is attending to the right features, and how to select an architecture that meets accuracy requirements within deployment constraints.

This paper documents the full research pipeline: dataset characterisation, preprocessing, systematic model comparison, final model selection, interpretability analysis, and deployment considerations. All code is available at github.com/345668/Plantrecognition---Computer-Vision.

2. Datasets

2.1. Dataset Selection

Three datasets were selected from Kaggle for inclusion in the study, based on the criterion that each must contain labelled health-status information (healthy vs. specific disease type) across mature plant leaves at consistent resolution:

Dataset	Description
PlantVillage Dataset	~55,000 images in three formats: colour, grayscale, segmented. 38 plant-health classes. Primary training dataset. Source: abdallahalidev/plantvillage-dataset
New Plant Diseases Dataset	70,297 training images, 17,573 validation images, 33 test images. Largest available; used for cross-dataset validation. Source: vipoooool/new-plant-diseases-dataset
Plant Disease Dataset	43,457 training images, 10,850 test images. Similar class taxonomy to New Plant Diseases but smaller scale. Source: saroz014/plant-disease

A fourth dataset, V2 Plant Seedlings, was evaluated and excluded. It lacks health-status labels, contains seedling stages not represented in the other datasets, and shows inconsistent image dimensions, making it unsuitable for joint training with the three selected datasets.

2.2. Dataset Characteristics and Class Distribution

All three selected datasets use a consistent 256×256 pixel resolution for the majority of images. The split between training and test sets varies significantly: New Plant Diseases allocates an unusually large validation set (17,573 images) relative to its test set (33 images), suggesting a focus on extensive validation during the original data collection.

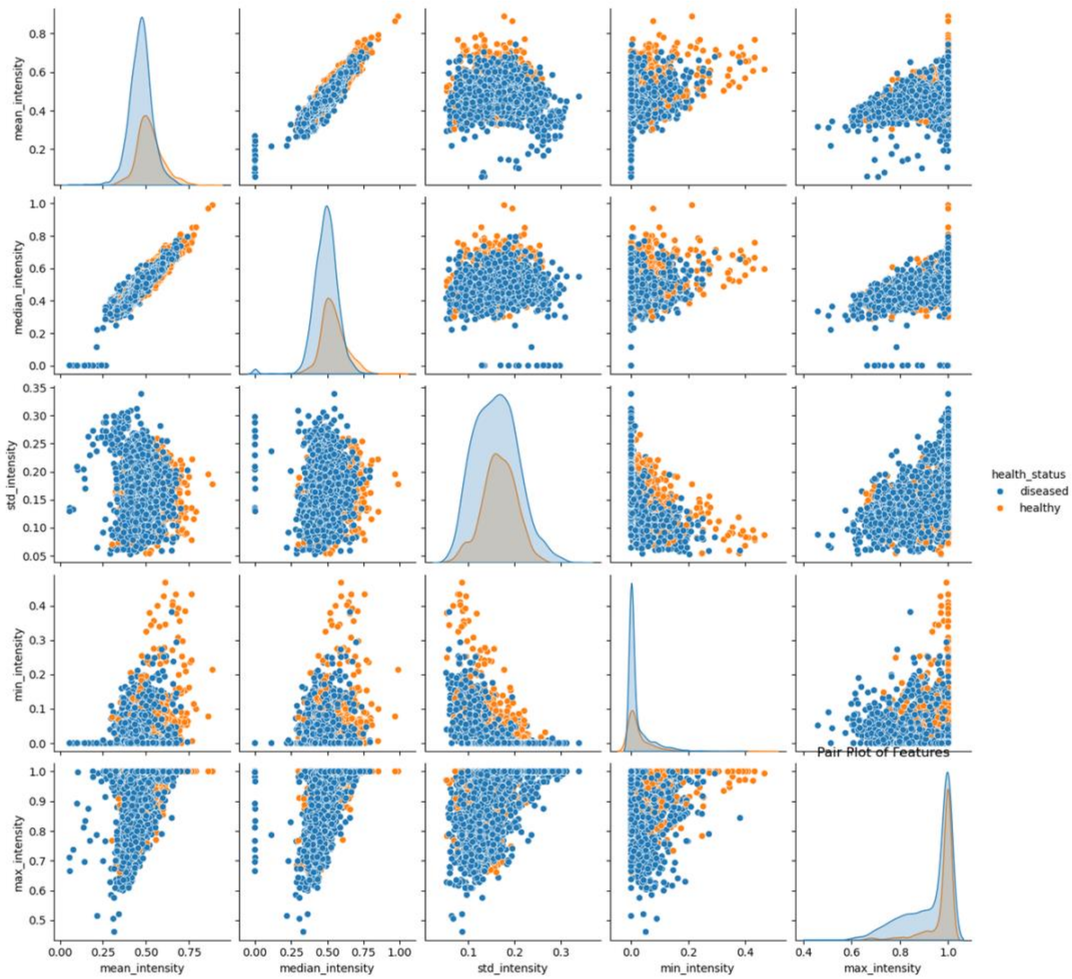

Figure 1. Image count distribution by plant category across three datasets. Tomato is significantly overrepresented; squash and raspberry are critically underrepresented.

2.3. Class Imbalance: Health Status

A consistent pattern across all three datasets is a disproportionate number of diseased images relative to healthy images. This imbalance is particularly pronounced for tomato, which dominates disease-class representation in the New Plant Diseases and Plant Disease datasets. The practical consequence is that a naive classifier trained on unbalanced data will achieve high apparent accuracy by over-predicting the disease state, while underperforming on healthy-plant detection.

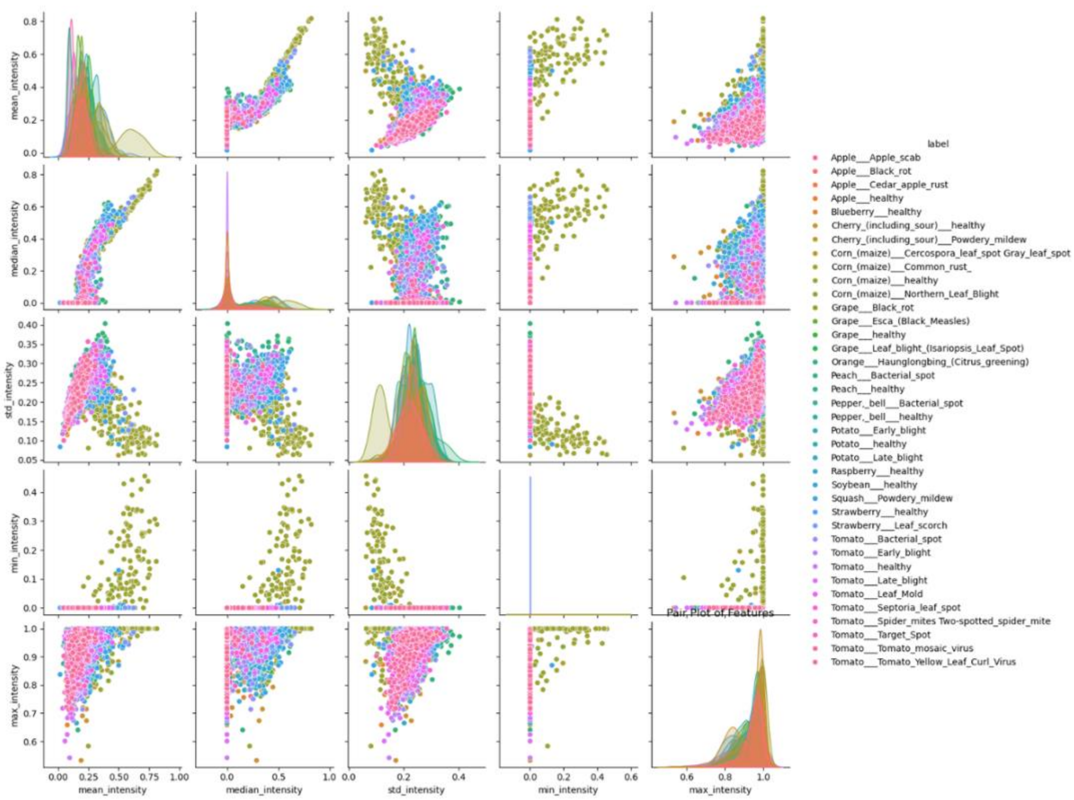

Figure 2. Overall distribution of healthy vs. diseased images across all three datasets. Diseased samples significantly outnumber healthy samples in all cases.

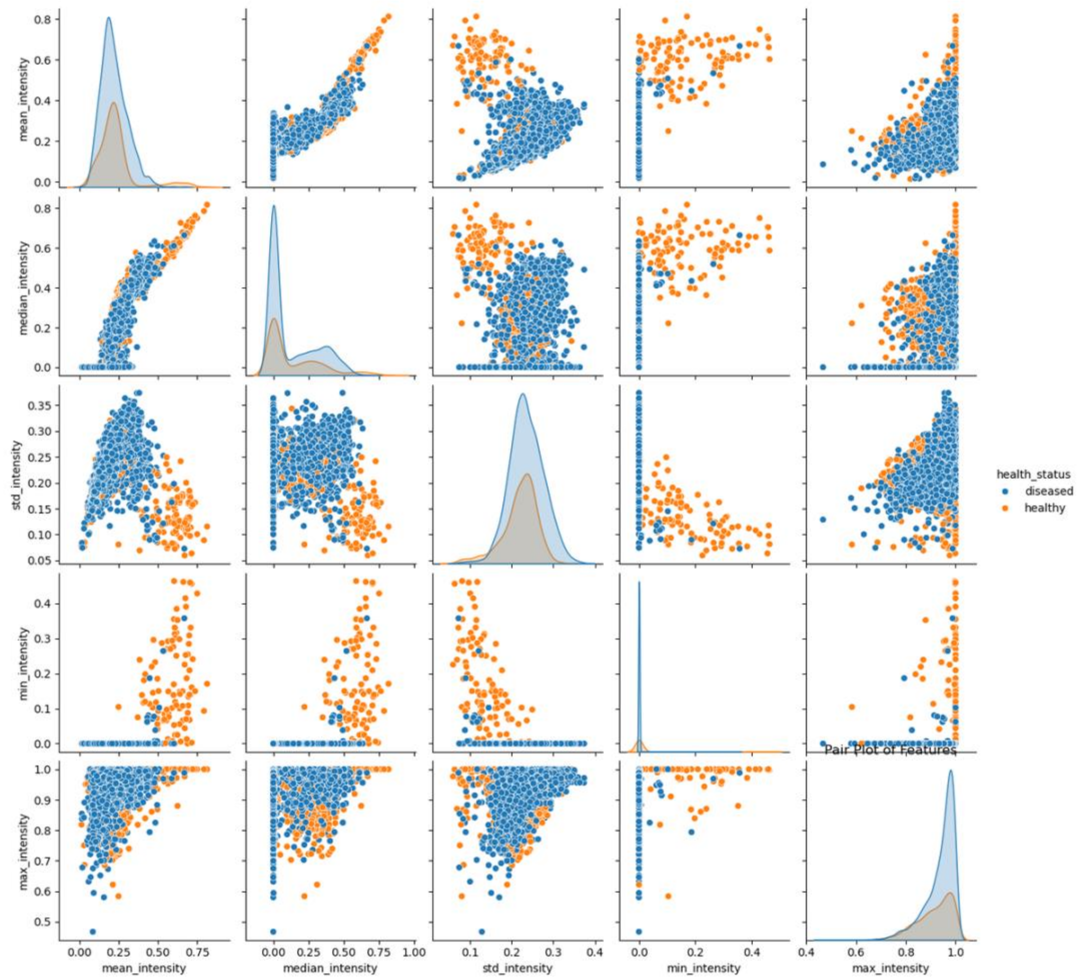

Figure 3. Health status distribution by plant category. Several species (squash, raspberry) have critical scarcity of healthy samples.

Key observations from the health status analysis:

- Tomato: nearly zero healthy samples relative to diseased. Creates a high overfitting risk toward tomato disease patterns.
- Apple, Blueberry, Corn: significantly fewer healthy samples than diseased. Models trained on these distributions may generate false-positive disease detections.
- Squash, Raspberry: critically underrepresented in both states. Models are unlikely to perform reliably for these crops without targeted augmentation.

2.4. Disease Distribution

The distribution of disease categories within datasets is non-uniform. Apple Scab and Cedar Apple Rust are overrepresented in the New Plant Diseases dataset. Powdery Mildew and Leaf Mold show unbalanced distribution across datasets. Certain diseases appear only for specific plant types (e.g., Bacterial Spot only for Orange; Powdery Mildew only for Squash), creating sparsely populated class combinations.

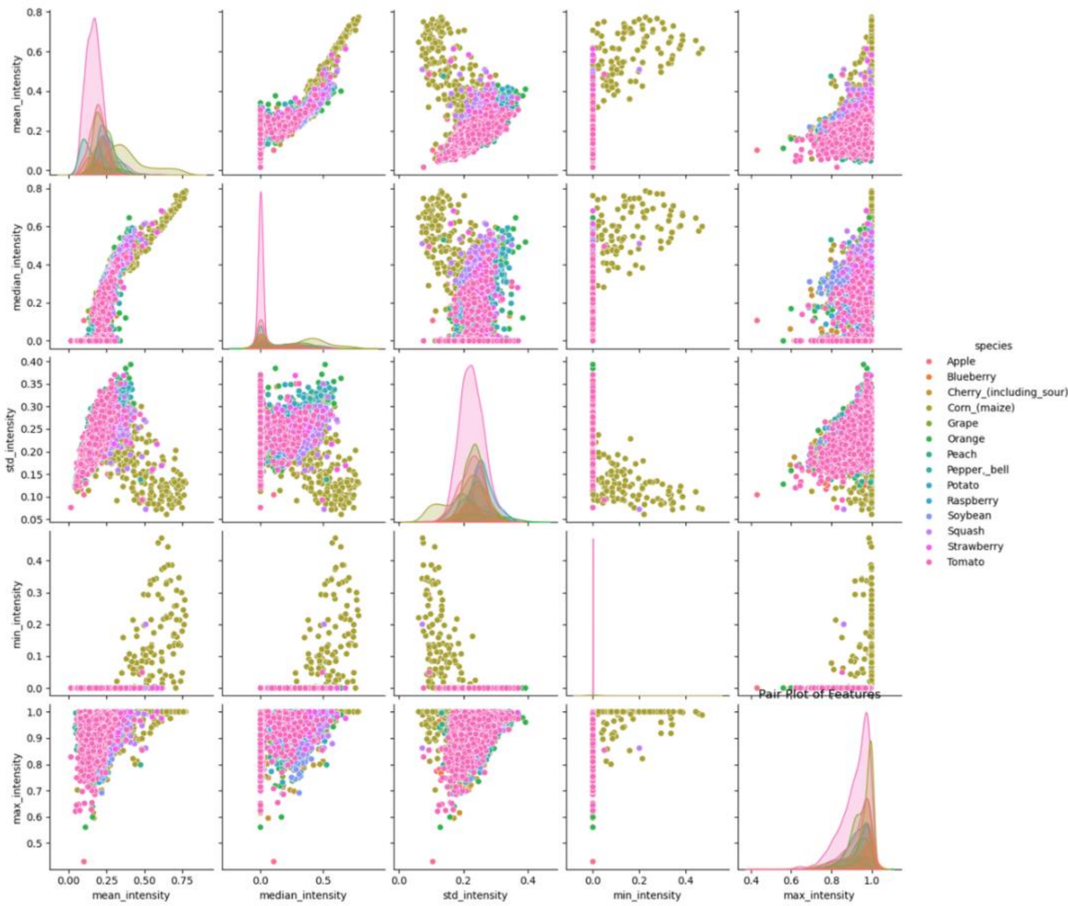

Figure 4. Disease-specific image count distribution across datasets. Several diseases are severely overrepresented; others have minimal sample counts.

3. Data Preprocessing

3.1. Dataset Combination and Scaling

For the primary training pipeline, we merged the colour and segmented subsets of the PlantVillage dataset, producing approximately 360,000 training images and ensuring balanced class representation across both image types. Due to hardware constraints (GPU memory limits on the DataScientest training infrastructure), all images were downscaled to 64×64 pixels for classical ML models and the initial CNN, and to 256×256 pixels for the VGG16 transfer learning phase.

3.2. Segmentation

Segmented images — where the leaf is isolated from the background — proved to be a significant preprocessing benefit. Background pixels introduce irrelevant colour information that distorts histogram-based features and can mislead texture feature extractors. The segmented PlantVillage subset provides a direct comparison between unsegmented and segmented versions of the same leaves.

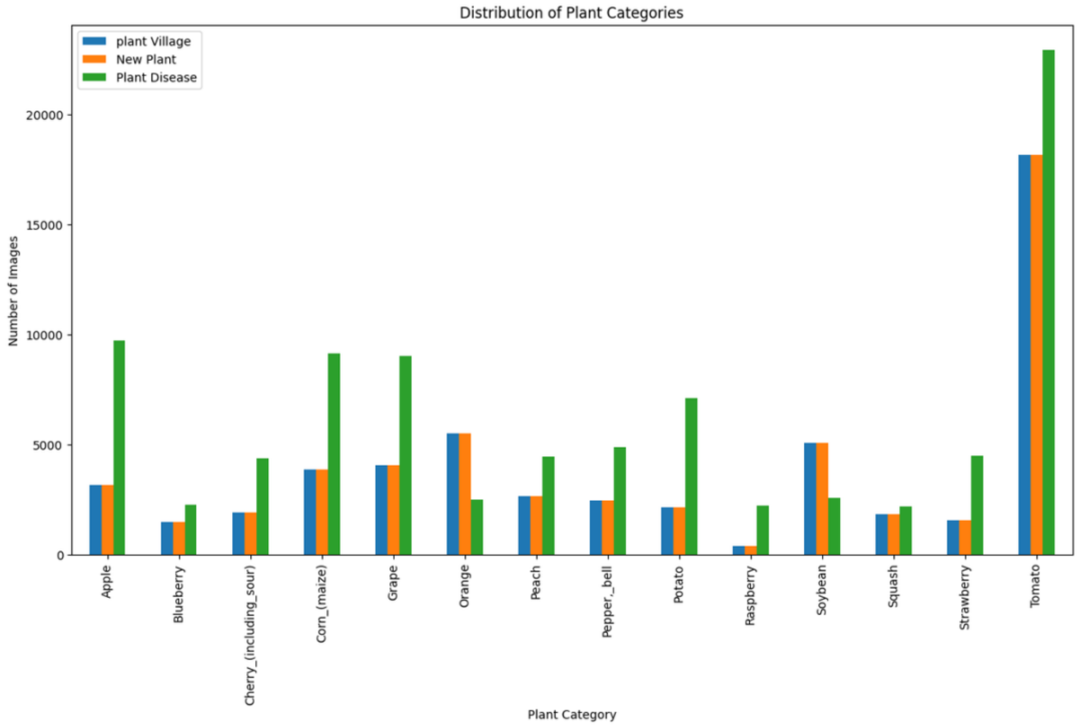

Figure 5. Colour distribution histograms for healthy (left) and diseased (right) leaves. Non-segmented images show substantial background contamination in the red and blue channels.

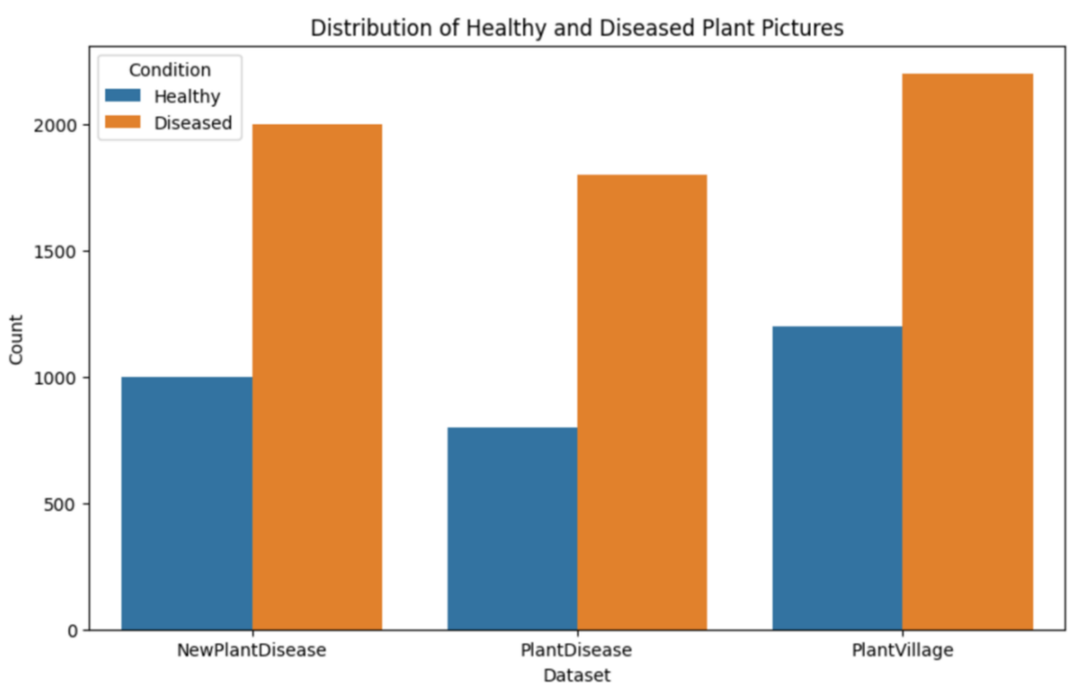

Figure 6. Segmented vs. non-segmented leaf colour distributions. Segmentation reveals markedly cleaner disease-specific colour signatures.

The empirical result from feature extraction experiments confirms this: pixel intensity scatter plots show substantially tighter, more separable clusters for segmented images compared to unsegmented images across both species and health-status classification tasks.

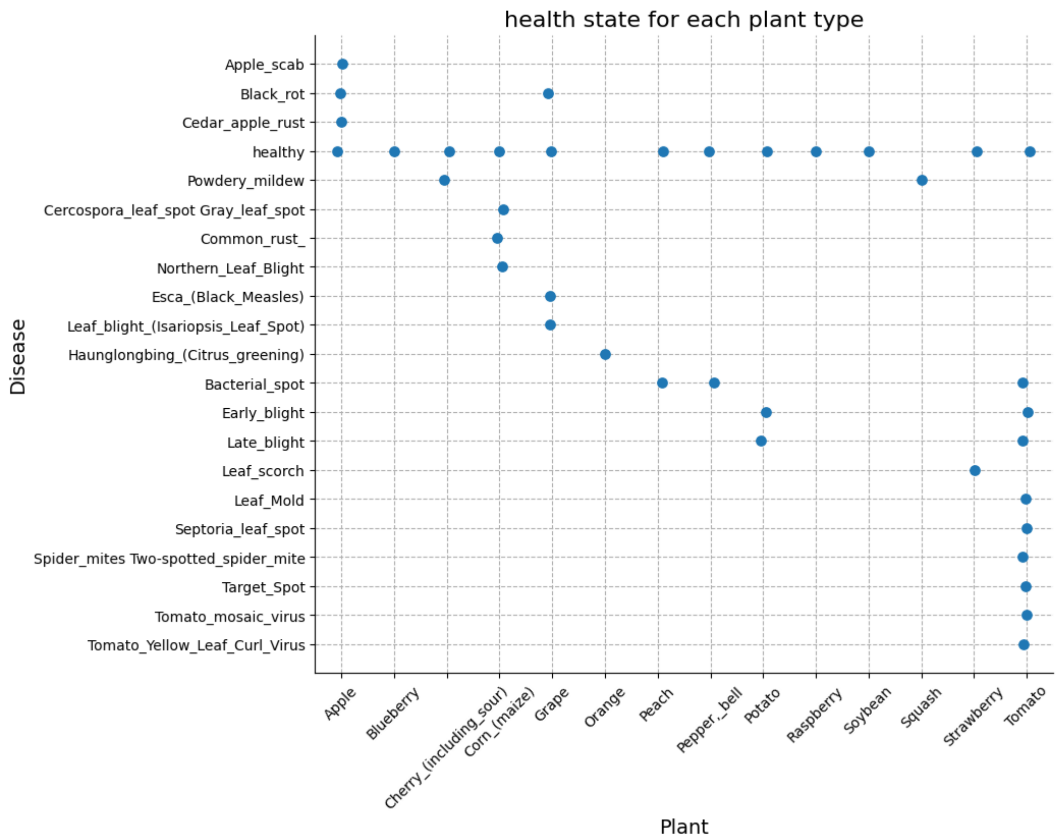

Figure 7. Pixel intensity scatter plots for species classification. Segmented images (right) show noticeably tighter species clusters versus non-segmented images (left).

3.3. Data Augmentation

To address class imbalance and increase the effective training set size, data augmentation was applied throughout the pipeline. The augmentation parameters were:

- Rotation: 0° to 315° in configurable increments
- Zoom: up to 5%
- Width and height shifts: up to 10%
- Horizontal flipping
- Constant fill mode with black colour for border pixels

For VGG16 specifically, augmentation was applied via Keras' ImageDataGenerator with rotation, width/height shifts, zoom, and horizontal flips. Mixed precision training was enabled to reduce memory overhead and accelerate training on GPU hardware.

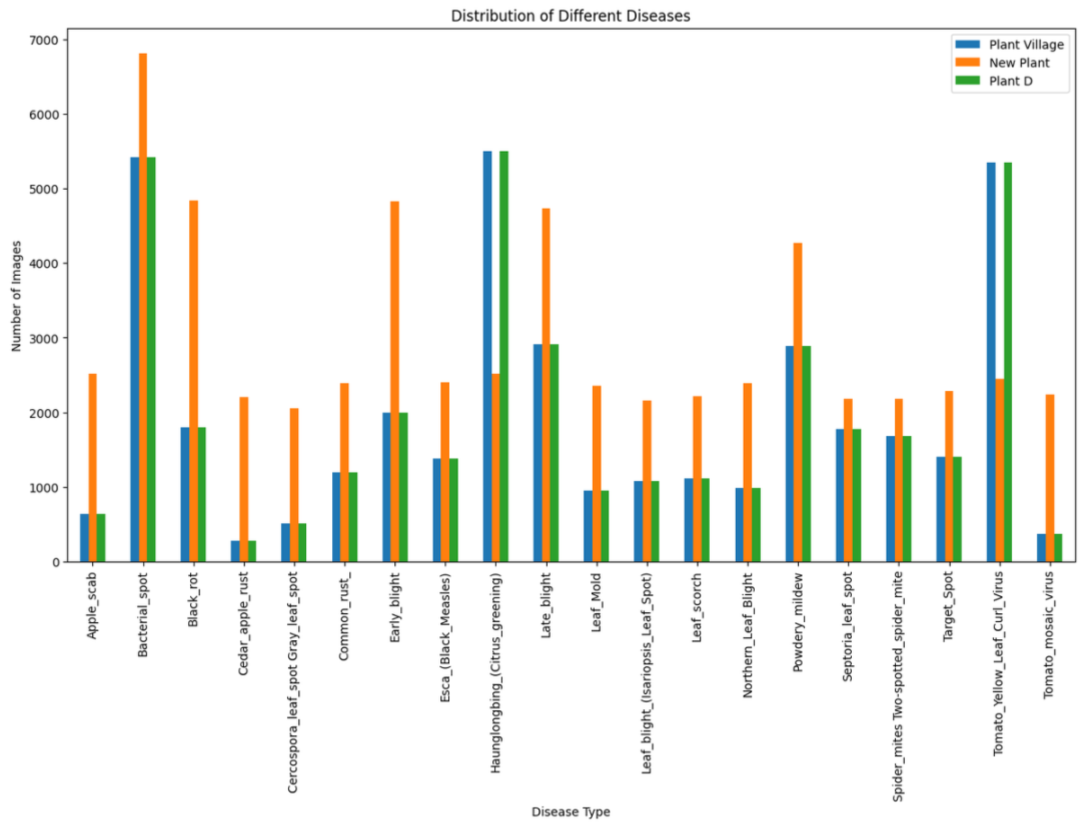

Figure 8. Examples of augmented training images. Rotations, flips, and zooms increase training diversity without requiring additional data collection.

3.4. Label Encoding and Splitting

Categorical class labels were encoded using scikit-learn's LabelEncoder for all models. One-hot encoding was applied for the CNN and VGG16 softmax output layers. All datasets were split 80/20 for training and testing. VGG16 used an additional validation split drawn from the training set for callback monitoring (early stopping, learning rate reduction on plateau, model checkpointing).

4. Model Selection and Evaluation

4.1. Classical Machine Learning Baseline

Before advancing to deep learning, a systematic evaluation of classical ML approaches was conducted to establish a baseline and to understand the feature landscape. Seven model configurations were tested:

4.1.1. Random Forest + PCA

Hyperparameter tuning selected 250 estimators as optimal. Overall accuracy: 48% (disease classification), 50% (plant type classification). Disease classification performance was higher for visually distinctive classes (Cercospora leaf spot) and lower for classes with visual similarity (Black rot, Cedar apple, Leaf blight). The 48% weighted F1-score indicates moderate but insufficient classification capability for deployment.

4.1.2. HOG + PCA + Decision Tree

Three-stage pipeline: plant type classification (accuracy 0.28), healthy vs. diseased (0.59), disease classification (0.21). The low plant type accuracy reflects the high visual similarity between species in leaf-only images. The moderate healthy vs. diseased accuracy suggests that coarse health signal is present in HOG features, but disease-specific discrimination is not.

4.1.3. Gabor Filtering + LDA + Logistic Regression

Gabor filters capture spatial frequency and texture at multiple orientations and scales — properties directly relevant to the leaf surface texture changes associated with fungal and bacterial disease. Combined with LDA dimensionality reduction and optimised Logistic Regression (Grid Search), this pipeline achieved 73.6% overall accuracy — the best classical ML result.

High precision and recall were achieved for visually distinctive healthy classes (Blueberry healthy, Corn healthy, Grape healthy). Lower performance was observed for diseases with subtle texture changes (Tomato Early Blight, Tomato Target Spot), reflecting the limits of hand-engineered texture features.

4.1.4. LBP + k-Nearest Neighbours

Local Binary Patterns with 24 neighbours and radius 3 (P24_R3) achieved the best LBP configuration with approximately 50% accuracy. The optimal k=17. Key observation: the LBP method effectively captures texture features, but the 50% ceiling indicates that LBP-derived features are insufficient for fine-grained disease discrimination.

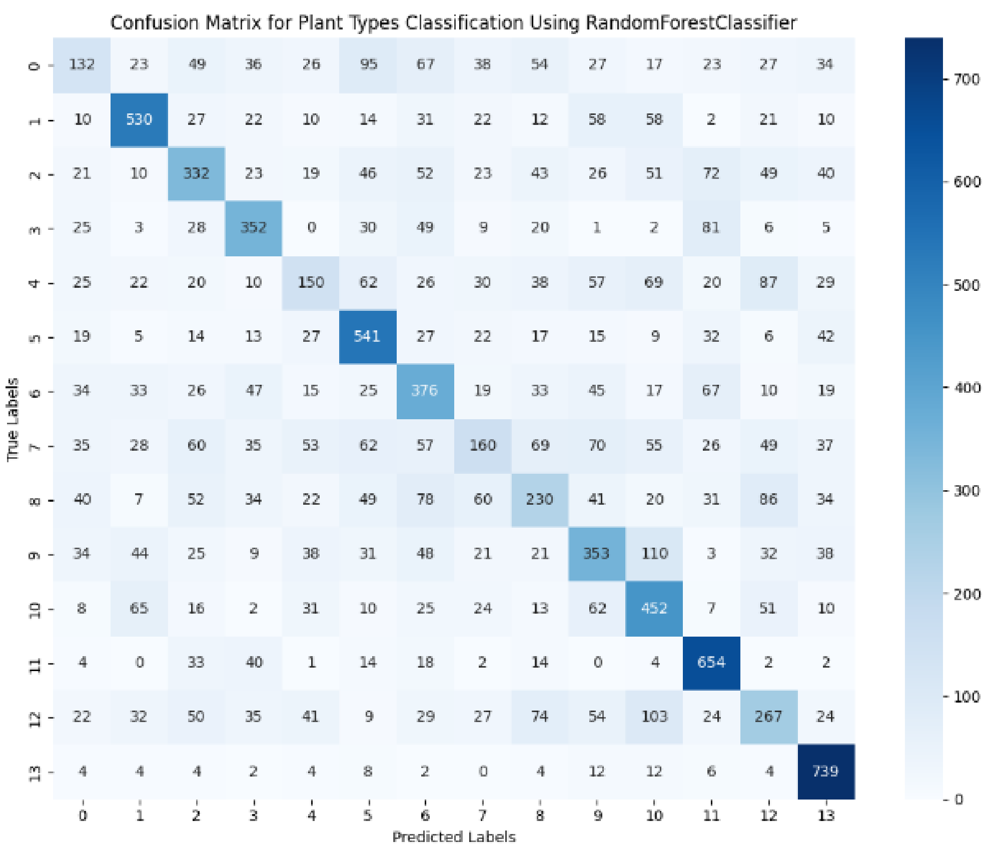

Figure 9. Accuracy across different LBP parameter configurations. P24_R3 achieves the best result at approximately 50%.

4.2. CNN (Custom Architecture)

The custom CNN architecture processes 64×64 RGB input through three shared convolutional layers (8, 16, 32 filters) with ReLU activation and MaxPooling, followed by a 128-neuron dense layer and 0.5 dropout. Separate output heads handle plant type and disease classification simultaneously (dual-branch architecture).

Training was conducted for 100 epochs with batch size 32. Test accuracy reached 73% on the disease classification task.

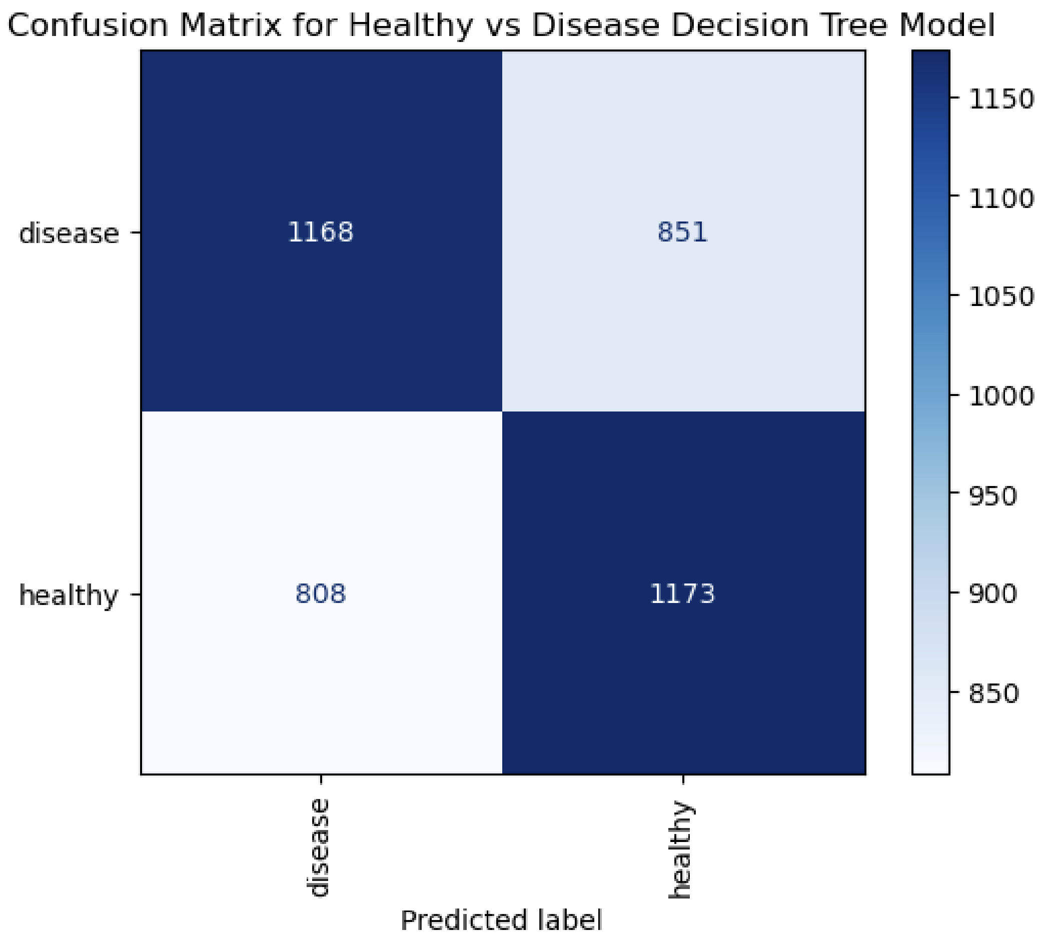

Figure 10. Custom CNN dual-branch architecture for simultaneous plant type and disease classification.

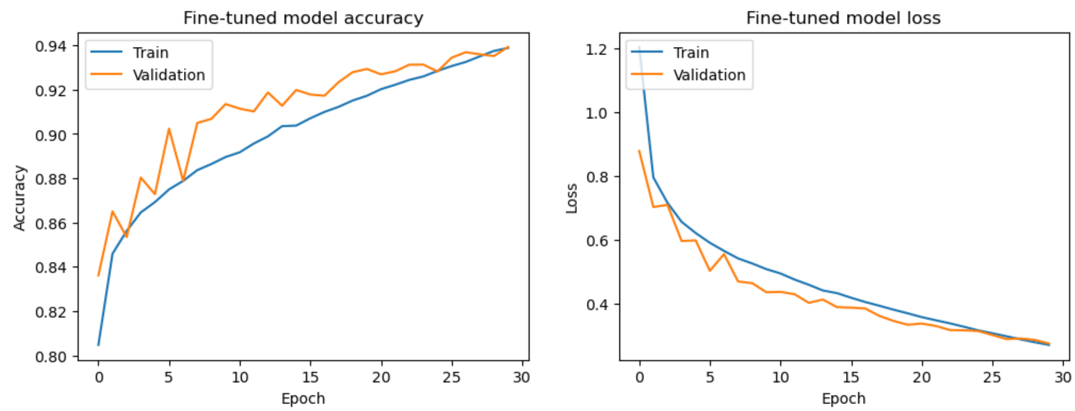

Figure 11. CNN training and validation accuracy curves for plant type (left) and disease (right) classification heads.

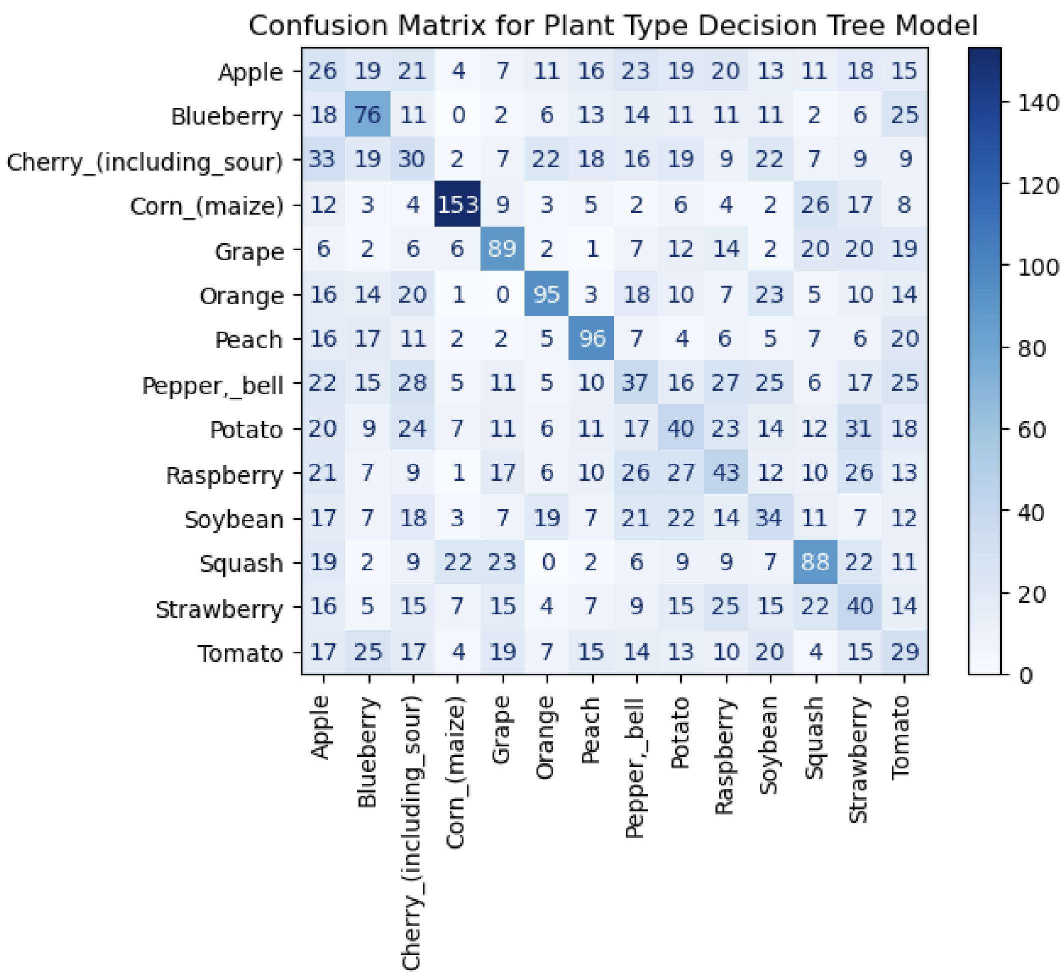

Figure 12. CNN training curves showing convergence. Validation accuracy tracks training accuracy without significant overfitting given the 0.5 dropout regularisation.

Critical limitation: Grad-CAM visualisation on the trained CNN revealed that the model frequently attended to background regions and image borders rather than leaf surface features, producing correct predictions for incorrect reasons. This is a standard failure mode for CNNs trained on small input resolutions (64×64) without background segmentation.

4.3. Transfer Learning with VGG16

VGG16 is a 16-layer convolutional neural network pre-trained on ImageNet (1.4 million images, 1000 classes). Its deep feature hierarchy — low-level edge detectors in early layers, complex texture and shape detectors in later layers — provides a strong prior for leaf surface analysis when fine-tuned on domain-specific data.

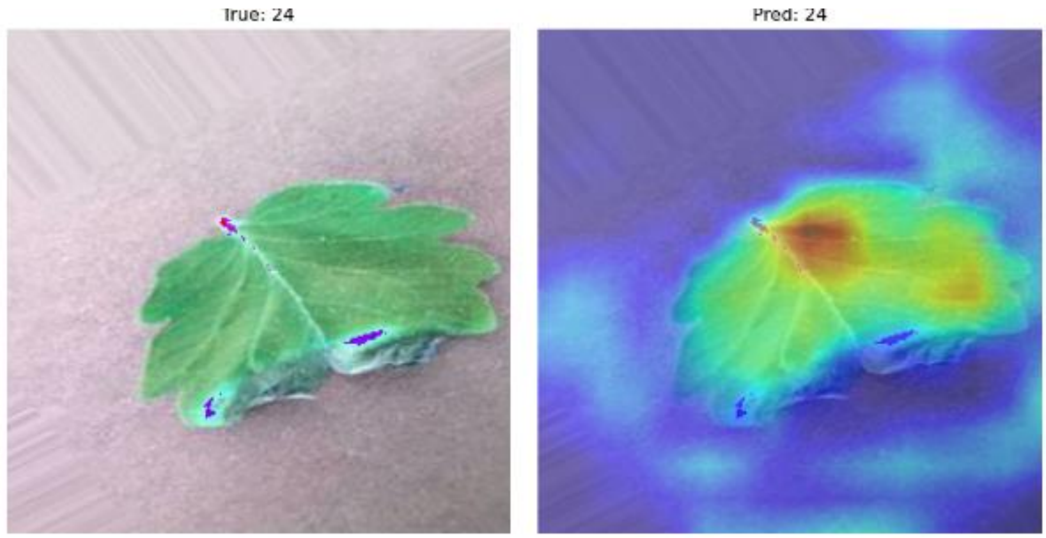

Figure 13. VGG16 architecture overview. Pre-trained convolutional layers are frozen; custom classification head is trained on PlantVillage data.

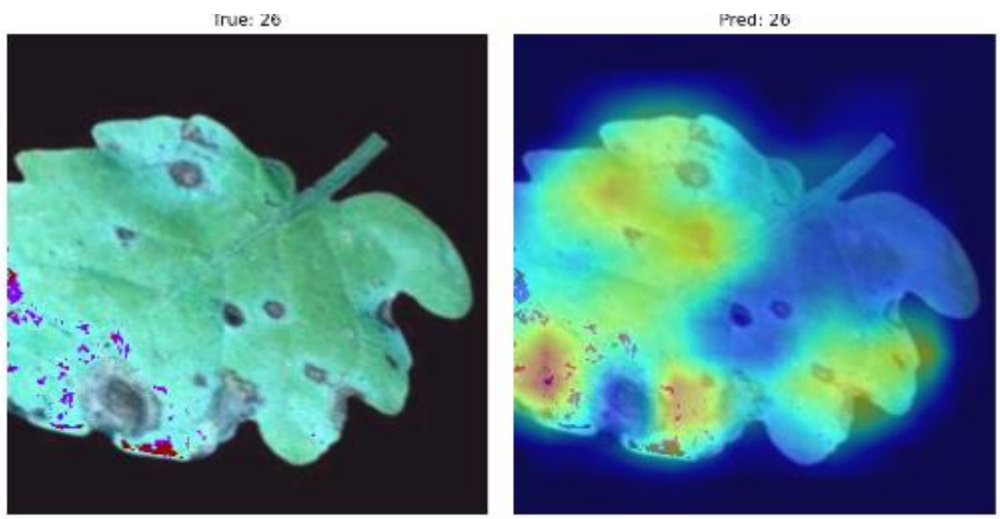

Figure 14. Transfer learning pipeline: ImageNet pre-trained weights → frozen VGG16 base → Global Average Pooling + Dense layers → 38-class SoftMax output.

4.3.1. Architecture Customisation

The standard VGG16 top classification layers were replaced with:

- Global Average Pooling layer (replacing the original 4096-unit dense layers)
- Dense layer with ReLU activation
- Dropout layer for regularisation
- Final Dense layer with SoftMax activation (38 output classes)

The VGG16 convolutional base was frozen during initial training. Mixed precision training (float16 forward pass, float32 weight updates) was enabled to fit the model within GPU memory constraints while maintaining numerical stability.

4.3.2. Training Configuration

- Optimiser: Adam with exponential learning rate decay
- Loss function: Categorical cross-entropy
- Callbacks: early stopping, learning rate reduction on plateau, model checkpointing (best validation accuracy)
- Batch size: 32 · Epochs: 15
- Input resolution: 256×256×3
- Training data: colour + segmented PlantVillage (~360,000 images with augmentation)

4.3.3. Grad-CAM Interpretability

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to validate that the model's predictions were driven by leaf surface features rather than background artefacts. For the VGG16 model, Grad-CAM consistently highlighted disease lesion areas, colour changes, and texture abnormalities on the leaf surface — confirming that the model had learned disease-relevant representations.

This was a critical qualitative validation step. The initial CNN failed this check (attending to borders); VGG16 passed it, providing confidence that the high accuracy metrics reflect genuine disease understanding rather than spurious correlation.

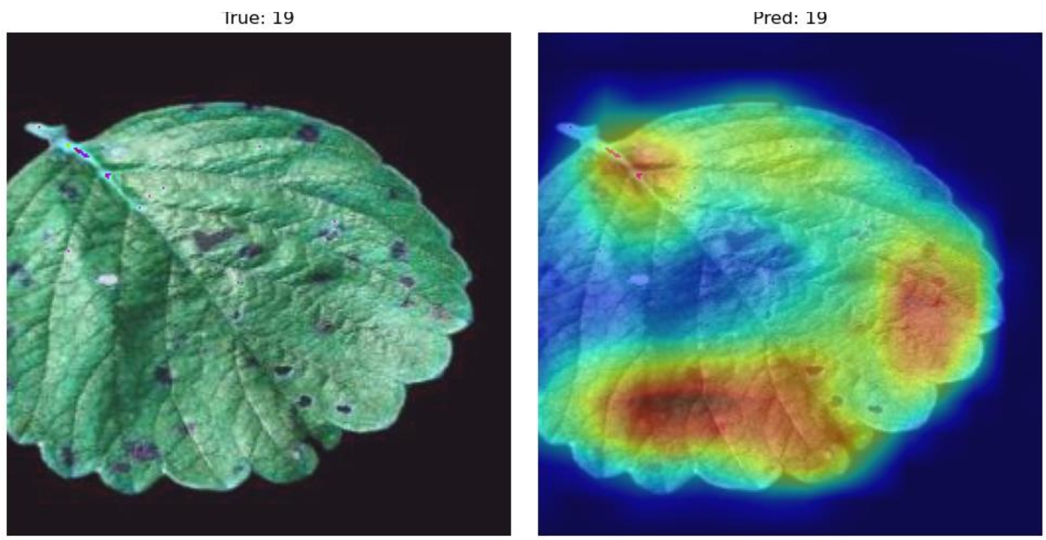

Figure 15. Grad-CAM visualisations for VGG16 on correctly classified diseased samples. Activation heat maps highlight disease lesion areas on the leaf surface.

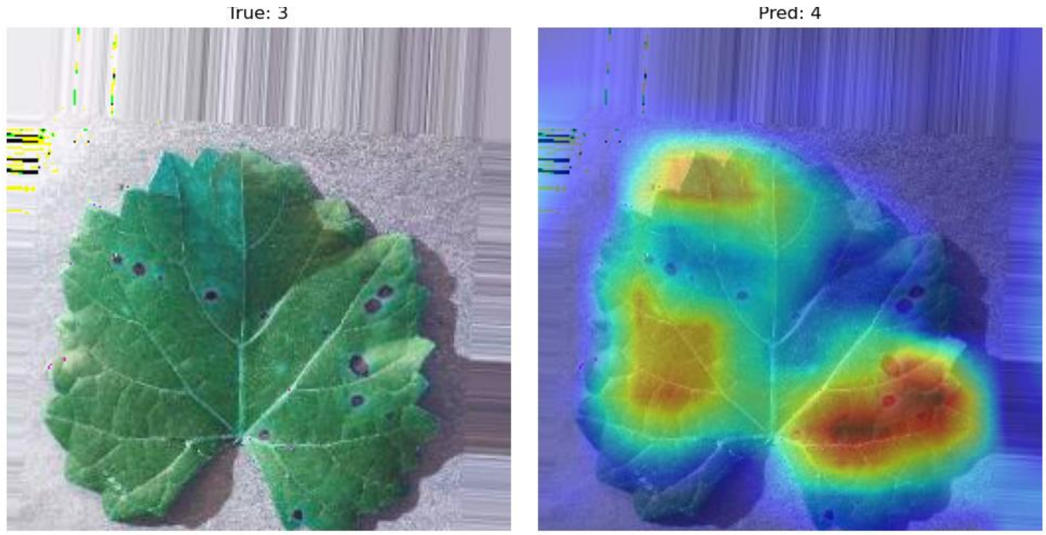

Figure 16. Additional Grad-CAM examples across plant species. The model consistently attends to disease-relevant regions across different plant types and disease presentations.

5. Results

5.1. Comparative Model Performance

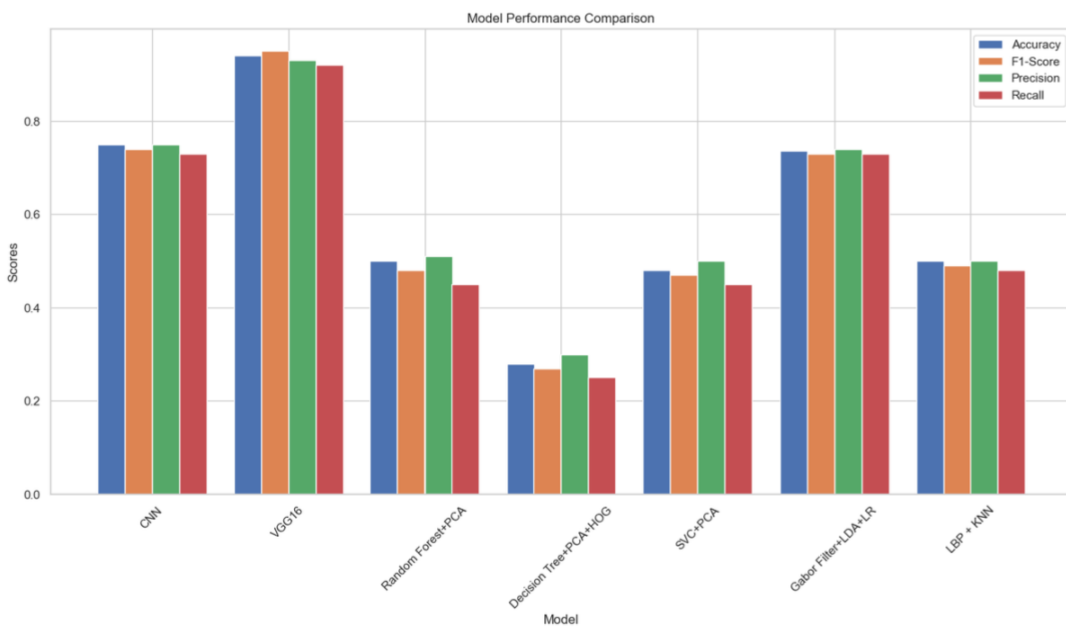

Figure 17. Comparative model performance across accuracy, F1-score, precision, and recall. VGG16 achieves best performance across all four metrics.

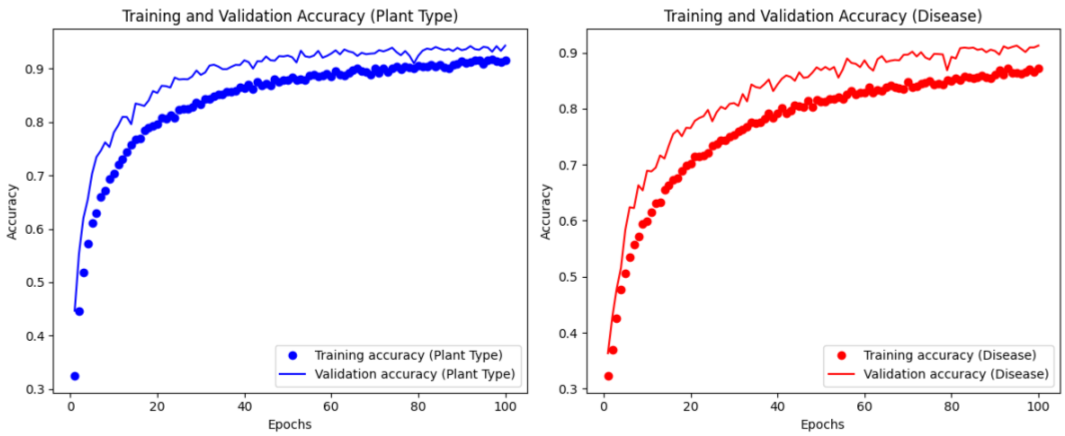

Figure 18. Detailed performance metrics for VGG16 vs. CNN. VGG16 shows substantial improvements in all evaluation dimensions.

Model	Features	Input	Dim. Red.	Accuracy	F1-Score	Train Time
PCA + SVM	None	64×64	PCA	—	—	17 min
HOG + PCA + DT	HOG	64×64	PCA	0.28–0.59	Low	9 min
PCA + Random Forest	None	64×64	PCA	0.48–0.50	0.48	7 min
LBP + k-NN	LBP	128×128	None	0.50	0.50	23 min
LDA + Log. Regression	None	64×64	LDA	—	—	20 min
Gabor + LDA + LR	Gabor	64×64	LDA	0.736	~0.73	13 min
Custom CNN	None	64×64	None	0.73	~0.73	983 min
VGG16 (Transfer)	None	256×256	None	0.94–0.96	0.95	943 min

5.2. VGG16 Final Model Performance

The optimised VGG16 model achieved the following metrics on the held-out test set:

- Test accuracy: 94–96% (reported consistently across evaluation rounds)
- F1-score: 95% (weighted across all 38 classes)
- Precision and recall: balanced, confirming no significant bias toward majority classes
- Inference time: 3 seconds per image on CPU; suitable for real-time mobile deployment

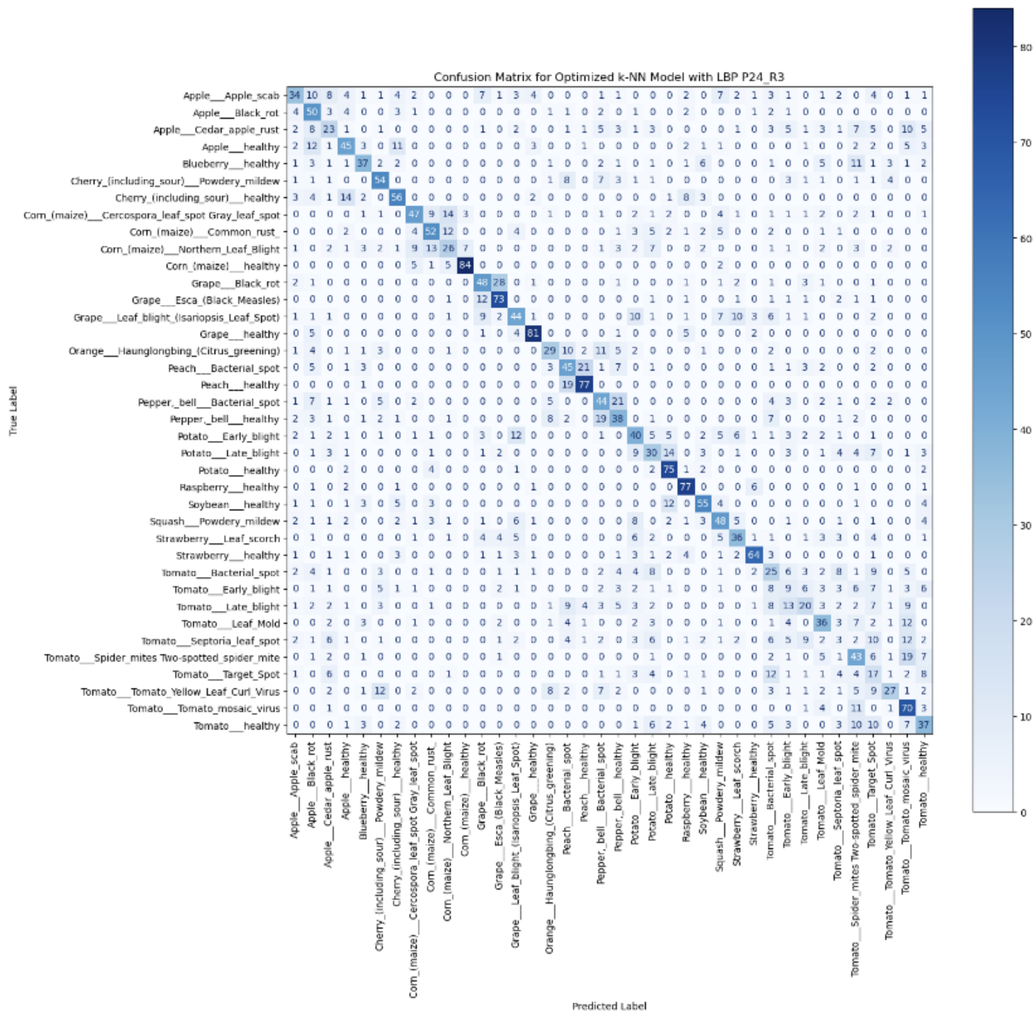

Figure 19. Confusion matrix for VGG16 on the test set. Strong diagonal dominance confirms reliable classification. Off-diagonal concentrations highlight visually similar disease pairs.

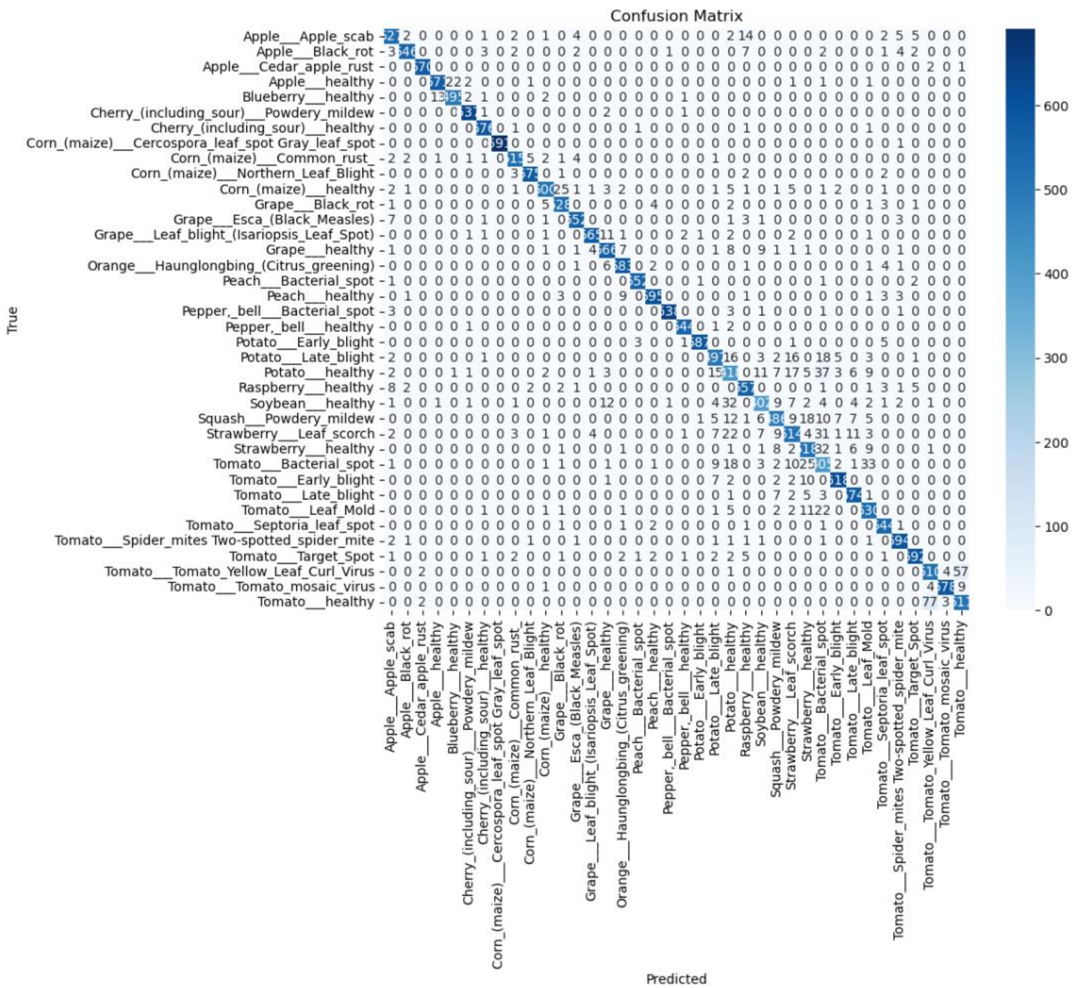

Figure 20. Per-class classification report for VGG16. High F1-scores across most of the 38 classes, with lower scores for visually similar disease conditions.

5.3. Key Performance Insights

Analysis of the confusion matrix and per-class classification report reveals three patterns:

- Visually distinctive healthy classes (Blueberry healthy, Corn healthy, Grape healthy) achieve near-perfect precision and recall.
- Diseases with highly characteristic visual signatures (Haunglongbing/citrus greening, Powdery mildew on grape) also achieve strong per-class scores.
- The most challenging confusions occur between diseases with similar visual presentations on different host plants (e.g., early blight on tomato vs. potato; bacterial spot variants), indicating that further improvement would require finer-grained feature discrimination or additional data.

5.4. Comparison with Baseline

The VGG16 model represents a 23–26 percentage point improvement in accuracy over the best classical ML model (Gabor+LDA+Logistic Regression at 73.6%) and a 21–23 percentage point improvement over the custom CNN (73% on 64×64 input). This improvement is attributable to three factors in roughly equal measure: the depth of the pre-trained feature hierarchy, the higher input resolution (256×256 vs. 64×64), and the background segmentation of training data that ensures feature learning focuses on leaf surface characteristics.

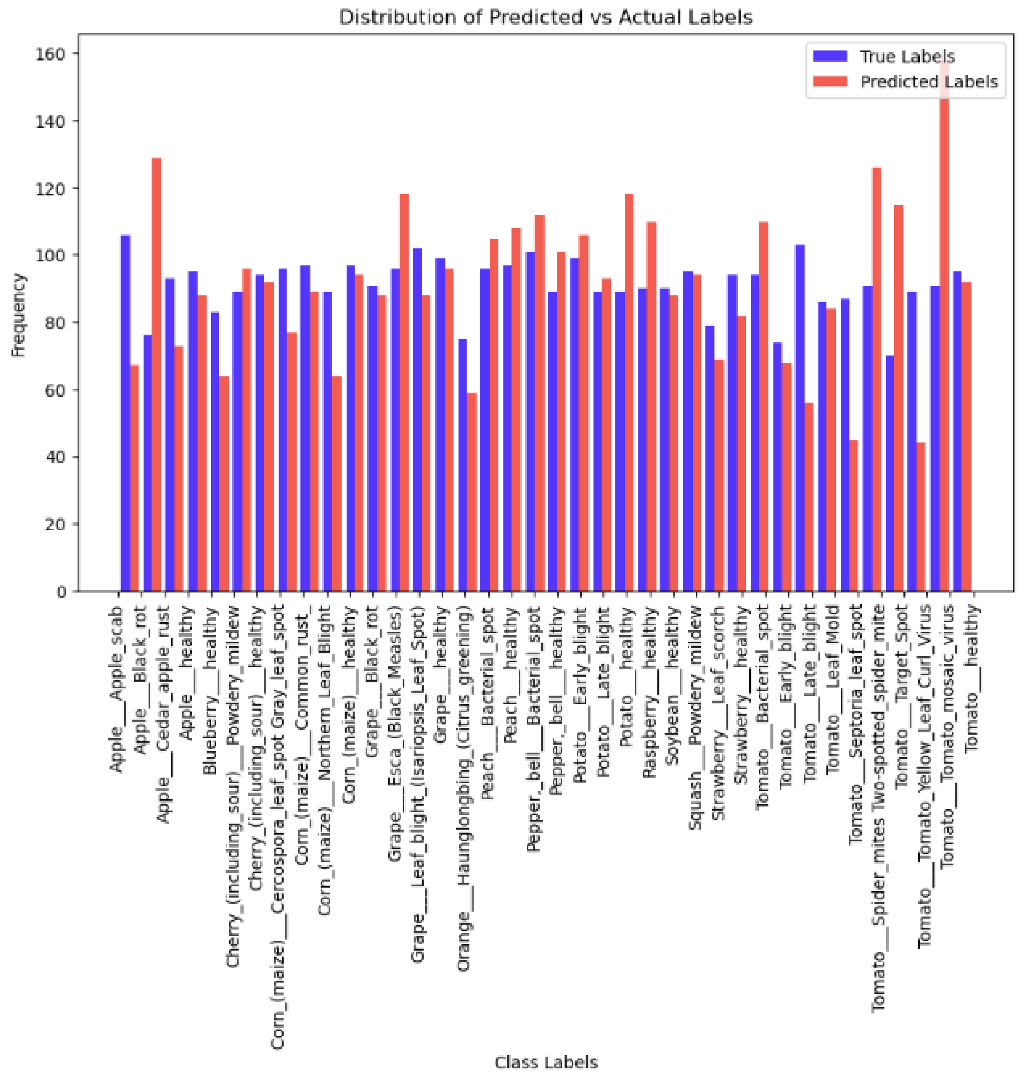

Figure 21. VGG16 training and validation accuracy curves over 15 epochs. Smooth convergence with minimal overfitting gap confirms effective regularisation.

6. Discussion

6.1. Why Transfer Learning Outperforms Classical Features

The performance gap between hand-engineered features (HOG, LBP, Gabor) and learned features (VGG16 convolutional layers) reflects a fundamental difference in what each approach captures. HOG captures gradient orientation histograms that encode coarse shape — useful for object detection but insufficient for fine-grained texture discrimination among 38 disease classes. LBP captures microstructure texture — closer to disease-relevant signal but limited to local contrast patterns. Gabor filters capture multi-scale spatial frequency — the best of the classical approaches, explaining its 73.6% accuracy ceiling.

VGG16's learned feature hierarchy captures all of these simultaneously and at multiple scales and compositions, without requiring the practitioner to specify which features are relevant. The critical enabler is the ImageNet pre-training: the network has already learned to detect edges, textures, materials, and shapes across 1,000 diverse categories. Fine-tuning on plant disease data specialises these representations for the target domain.

6.2. Role of Segmentation

The most consistent finding across both classical and deep learning experiments is the benefit of background segmentation. Non-segmented images introduce soil, sky, and support structure pixels that contribute to feature calculations without discriminative value. For classical methods (histogram-based features, colour distribution analysis), background pixels directly corrupt the feature vectors. For deep learning, background pixels provide misleading gradients that redirect feature learning away from leaf surface characteristics.

The segmented PlantVillage subset resolved this cleanly. For deployment scenarios where segmentation is not pre-applied (e.g., field photography with smartphones), a preprocessing step — either a simple green-channel thresholding mask or a lightweight segmentation model — is a prerequisite for reliable disease detection.

6.3. Class Imbalance Implications

The tomato-dominated class distribution required active mitigation via RandomUnderSampler and augmentation. Without balancing, models trained on raw class distributions achieve high apparent accuracy by overpredicting common tomato disease classes while failing on underrepresented crops. This is a structural problem of the available datasets that reflects the agricultural research community's focus on economically significant staple crops.

For production deployment across a diverse crop portfolio, additional data collection for underrepresented species (squash, raspberry) and more balanced sampling strategies would be necessary.

6.4. Grad-CAM as a Required Validation Step

A model achieving high aggregate accuracy is not necessarily a reliable model. The initial CNN case demonstrates this: 73% test accuracy with Grad-CAM showing attention to background and border regions. Had Grad-CAM not been applied, this model might have been deployed with a fundamental structural flaw — correct predictions for incorrect reasons, with brittle behaviour on images with different backgrounds or framing.

For any plant disease classification system intended for field use, Grad-CAM or equivalent saliency analysis should be treated as a required validation step, not an optional interpretability exercise.

6.5. Computational Constraints

VGG16 required 943 minutes of training on the DataScientest GPU infrastructure. This training time reflects the scale of the augmented dataset (~360,000 images) and the 256×256 input resolution. For practitioners without access to GPU infrastructure, mixed precision training and learning rate scheduling are essential for making VGG16-scale training feasible. Lighter architectures (MobileNetV2, EfficientNet-B0) could achieve comparable accuracy at significantly lower computational cost and should be considered for deployment on edge devices.

7. Challenges and Future Work

7.1. Remaining Challenges

- Visually similar disease pairs: early blight patterns on tomato and potato; bacterial spot variants across species. Finer feature discrimination is required.

- Dataset geographic bias: all three datasets are laboratory-controlled images. Field photography under variable lighting, occlusion, and angle produces significant domain shift.
- Rare species coverage: squash, raspberry, and several other crops have insufficient training examples for reliable deployment.
- Computational cost: 943-minute training time is impractical for iterative development; cloud-based GPU infrastructure or lighter architectures are required.

7.2. Prioritised Future Directions

- Real-world data collection: field images under natural lighting, multiple angles, and partial leaf occlusion. Diversity of source environments (equatorial, temperate, greenhouse) is essential for generalisation.
- Lightweight architecture evaluation: MobileNetV3 and EfficientNet-B0 offer comparable accuracy to VGG16 at 5–10× lower inference time, making them better candidates for mobile and edge deployment.
- Expanded disease taxonomy: integration with agricultural disease databases to support detection of diseases not represented in PlantVillage (e.g., soil-borne pathogens, viral infections visible at early stages).
- Environmental variable integration: combining leaf imagery with sensor data (soil moisture, temperature, humidity) to improve disease prediction before visible symptoms appear.
- Mobile and robotic deployment: wrapping the trained VGG16 model in a mobile API with a camera interface, enabling real-time field diagnosis from a smartphone or drone-mounted camera.
- Collaborative agricultural validation: partnership with agronomists and plant pathologists to validate model outputs against expert diagnoses in production settings.

8. Conclusion

This paper presented a systematic evaluation of computer vision approaches for plant disease classification, progressing from classical machine learning baselines through a custom CNN to transfer learning with VGG16 on the PlantVillage dataset.

The central finding is that transfer learning with VGG16, combined with background-segmented training data and class-balanced augmentation, achieves 94–96% test accuracy and a 95% F1-score on 38 plant health classes — a 23+ percentage point improvement over the best classical ML approach and a 21+ percentage point improvement over the custom CNN. Grad-CAM analysis confirmed that this accuracy reflects genuine disease-specific feature learning rather than spurious pattern matching.

The practical significance of this work lies not in the accuracy figure itself — VGG16-level performance on PlantVillage has been reported elsewhere — but in the comparative pipeline: the rigorous evaluation of why classical methods fall short, the identification of segmentation and class-balanced augmentation as the highest-leverage preprocessing steps, and the use of Grad-CAM as a required (not optional) validation mechanism.

The resulting system is architecturally ready for integration into precision agriculture platforms and robotic systems. Real-world deployment will require field data collection to address domain shift, and likely a transition to lighter architectures (MobileNet, EfficientNet) for the computational constraints of edge devices. These are engineering challenges, not fundamental barriers — the core finding stands: deep learning-based plant disease detection at deployment-grade accuracy is achievable with publicly available datasets and standard compute infrastructure.

Appendix A: Software Environment

Component	Version / Notes
Python	3.x — primary implementation language
TensorFlow / Keras	Model training, ImageDataGenerator, callbacks
VGG16 base	Pre-trained ImageNet weights via keras.applications
scikit-learn	Classical ML models, PCA, LDA, LabelEncoder, RandomUnderSampler
OpenCV	Image loading, colour space conversion, preprocessing
Matplotlib / Seaborn	Visualisation of distributions, confusion matrices, training curves
Grad-CAM	Layer-wise gradient visualisation for model interpretability
NumPy / Pandas	Numerical operations, data management
Kaggle Datasets	PlantVillage, New Plant Diseases, Plant Disease (public)

Appendix B: Dataset Summary Statistics

Dataset	Key Statistics
PlantVillage Dataset	~55,000 images · 38 classes · Formats: colour, grayscale, segmented · 256×256px · Primary training source
New Plant Diseases Dataset	Train: 70,297 · Validation: 17,573 · Test: 33 · Large validation set for class-level model tuning
Plant Disease Dataset	Train: 43,457 · Test: 10,850 · Standard 80/20 split · Same taxonomy as New Plant Diseases
Combined (colour + segmented PlantVillage)	~360,000 images after augmentation · Used for VGG16 fine-tuning · Class-balanced via RandomUnderSampler
Input resolution (ML models)	64×64 pixels (hardware constraint during training)
Input resolution (VGG16)	256×256 pixels
Train/test split	80/20 across all models · Additional validation split for VGG16 callbacks

Appendix C: References

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DataScientest — Université Paris 1 Panthéon-Sorbonne, May 2024 Data Science Bootcamp. Project mentor: Roman Lesieur.